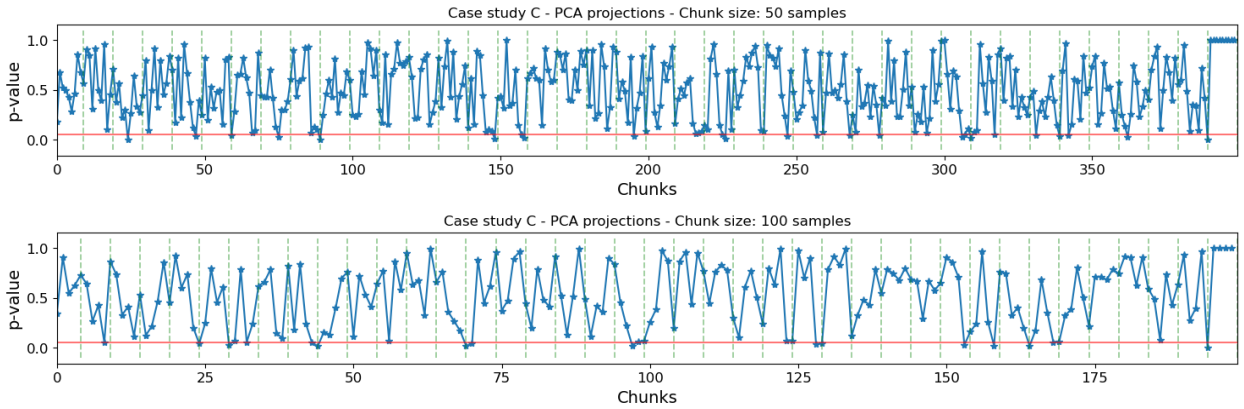


Figure 7: Results of the method using PCA projections in case study C for chunk sizes of 50 and 100 samples.

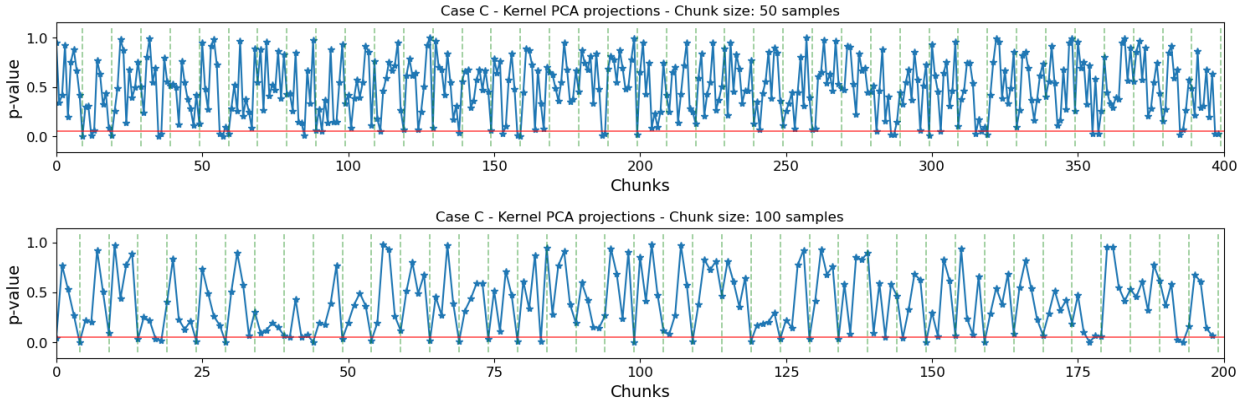


Figure 8: Results of the method using Kernel PCA projections in case study C for chunk sizes of 50 and 100 samples.

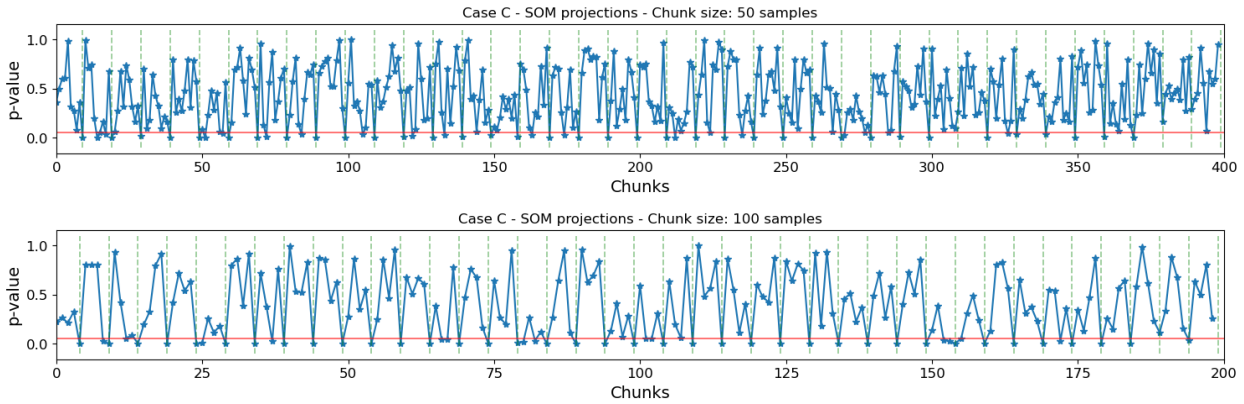


Figure 9: Results of the method using SOM projections in case study C for chunk sizes of 50 and 100 samples.

Table 1: Sensitive analysis of the chunk size for the three study cases, the three methods, and two levels of significance of the hypothesis test (0.05 and 0.1). The table shows the number of potential drifts (flags) detected by each method. The fourth column shows the number of injected drifts. This is an approximate number because, during the construction of the dataset, each image wasn't individually verified. Consequently, digits like 2, 4, 6, and 9 may sometimes have a hole and sometimes not.

| Case study | Chunk size | Method | Drifts | Flags (p-value at 0.05 level) | Flags (p-value at 0.1 level) |
|---|---|---|---|---|---|
| A | 50 | SOM | 20 | 13 | 30 |
| | | PCA | | 22 | 51 |
| | | KernelPCA | | 27 | 58 |
| | 100 | SOM | 20 | 9 | 16 |
| | | PCA | | 10 | 27 |
| | | KernelPCA | | 14 | 22 |
| | 250 | SOM | 20 | 3 | 5 |
| | | PCA | | 6 | 11 |
| | | KernelPCA | | 4 | 8 |
| B | 50 | SOM | 20 | 18 | 36 |
| | | PCA | | 17 | 34 |
| | | KernelPCA | | 17 | 39 |
| | 100 | SOM | 20 | 10 | 26 |
| | | PCA | | 6 | 12 |
| | | KernelPCA | | 10 | 21 |
| | 250 | SOM | 20 | 4 | 12 |
| | | PCA | | 3 | 4 |
| | | KernelPCA | | 9 | 14 |
| C | 50 | SOM | 40 | 53 | 75 |
| | | PCA | | 21 | 44 |
| | | KernelPCA | | 27 | 51 |
| | 100 | SOM | 40 | 43 | 53 |
| | | PCA | | 13 | 26 |
| | | KernelPCA | | 26 | 37 |
| | 250 | SOM | 40 | 40 | 43 |
| | | PCA | | 8 | 10 |
| | | KernelPCA | | 25 | 35 |

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